Constant Voltage PSR Controller with Active Power Factor Correction

Features

- High Precision Constant Voltage Regulation
- Primary Side Regulation Technology
- Constant On Time Control for Active PFC with High Power Factor (>0.9) and Low THD (<10%)
- Audio Noise Free Operation
- Fast Start-up
- Real Time Digital Compensating Technology for High Precision (±3%) Voltage Control, Excellent Line and Load Regulation
- Quasi-resonant Operation with Frequency Clamping for High Efficiency and Low EMI Operation
- Integrated 1KV SiC MOSFET
- HSOP7 Package

Protection

- Built-In Soft Start
- Output Open Protection
- Output Short (SCP)
- Over Temperature Protection (OTP)
- Over Load Protection (OLP)
- Cycle-by-cyble Current Limiting
- Leading Edge Blanking for Current Sensing
- VCC Under Voltage Lock Out (UVLO)
- VCC Over Voltage Protection (OVP)

Application

- AC-DC Adapter with PFC Function

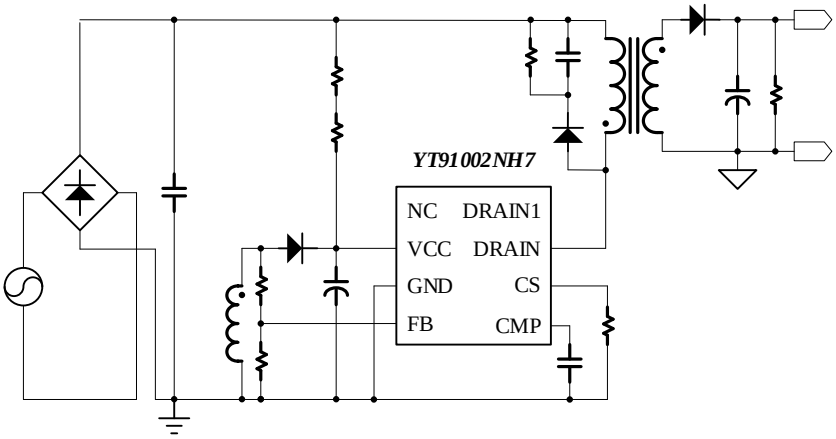


Fig. 1. Typical application circuit

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Package and Ordering Information

Part No.	Material Type	Package	Operating Temperature	Built-in MOSFET	Packing Method
YT91002NH7	Green	HSOP7	-40℃ to 105℃	1KV 0.48ohm	Tape 5000 pcs/Reel

Pin Configuration

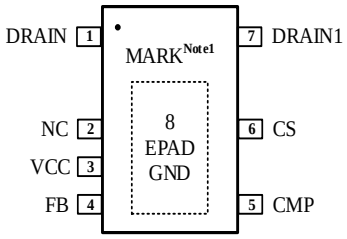


Fig. 2. Pin configuration

Note 1: Please refer to the "Marking Rule".

Pin Definition

Pin No.	Name	Description
1	DRAIN	Drain of the integrated high voltage MOSFET
2	NC	No connection
3	VCC	Power supply buffer for chip
4	FB	Voltage feedback from auxiliary winding with voltage divider for output voltage control
5	CMP	Loop compensation with a capacitor connected to GND
6	CS	Current sense for maximum loading setting with a resistor connected to GND
7	DRAIN1	The drain of the internal LV silicon MOS and same as source of the SiC MOS. No external connection is required, keep floating
8/EPAD	GND	Ground

Internal Block Diagram

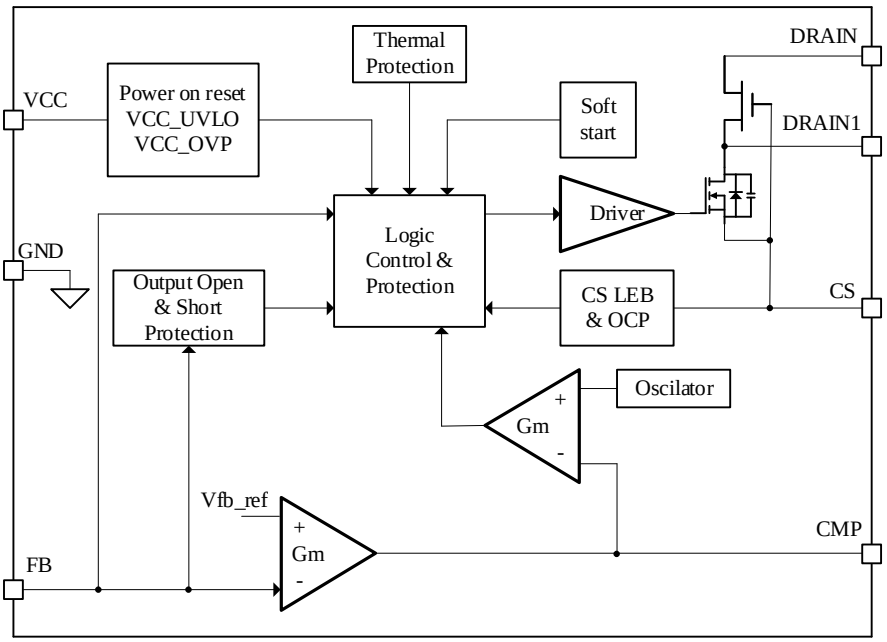


Fig. 3 Internal block diagram

Absolute Maximum Ratings[Note2]

Parameters	Symbol	Range	Units
"VCC" voltage range	V_{VCC}	-0.3 ~ 45	V
"CS", "FB", "CMP" voltage range	V_{CS}, V_{FB}, V_{CMP}	-0.3 ~ 7	V
Guaranteed minimum "DRAIN" voltage[Note3]	V_{DS_min}	1000	V
Guaranteed minimum "DRAIN1" voltage[Note3]	V_{DS1_min}	30	V
ESD Human mode[Note4]	ESD_{hbm}	2000	V
Operating junction temperature range	T_j	-40 ~ 150	℃
Ambient temperature range	T_a	-40 ~ 105	℃
Storage Temperature Range	T_{stg}	-65 ~ 150	℃
Welding temperature (< 20 s welding)	T_{lead}	260	℃
Junction to ambient thermal resistance	θ_{thja}	45	℃/W

Note 2: Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device.

Note 3: Depending on the different built-in MOSFET SPEC, see the corresponding relationship between the model and the built-in MOSFET.

Note 4: Electrical components and circuit boards will be aware of the situation in the discharge is not easy. Although this product has a special electrostatic protection circuit, but in the case of high-energy electrostatic discharge, the chip may have damage and loss of function or performance degradation. Therefore, users still need to take appropriate preventive measures ESD.

Electrical Characteristics[Note5] (Unless otherwise specified, T_a = 25 ℃)

Parameter	Symbol	Conditions	Min	Typ.	Max	Units
VCC						
VCC Start-up Current	I_{ST}	V_{VCC} = V_{UVLO_OFF} -1 V	3	7	20	μA
VCC Operation Current	I_{OP}	V_{VCC} = 20V	0.8	1	1.4	mA
VCC Under Voltage Lock Out Enter	V_{UVLO_ON}	V_{VCC} Falling	5.8	6.8	7.8	V
VCC Under Voltage Lock Out Exit	V_{UVLO_OFF}	V_{VCC} Rising	20	22.5	25	V
VCC Over Voltage Protection	V_{VCC_OVP}		29	32	35	V
PWM						
Maximum On Time	T_{on_max}			22		μs
Maximum Off Time	T_{off_max}			75		μs
Maximum Frequency	f_{max}			100		kHz
Minimum Frequency	F_{min}			500		Hz
CS						
CS Sampling OCP Voltage	V_{CS_OCP}	T_{ON} = 0 μs		0.7		V
CS Leading Edge Blanking Time	T_{LEB_CS}			300		ns
FB						
Output OVP Threshold	V_{OVP_FB}		3	3.2	3.4	V
FB Short Protection	V_{SCP_FB}			0.4		V
Gm Amplifier						
Reference Voltage	V_{REF}		2.425	2.5	2.575	V
Transconductance Gain	Gm			40		μS
MOS						
Drain-source breakdown voltage	BV_{DSS}	Drain to source voltage @V_{GS}=-30V	1000			V
Drain-source on resistance	R_{DS_ON}	V_{GS}=12V，I_D=2A，T_J=25℃		0.48		ohm
OTP						
OTP Threshold	T_{OTP}			150		℃

Note 5: The max and minimum parameters specified are guaranteed by test, the typical value are guaranteed by design, characterization and statistical analysis.

Application Suggestion

YT91002NH7 is a high precision active PFC controller, specially designed for PSR flyback offline constant voltage control. Operating in critical conduction mode with quasi-resonant to achieve high power factor, low THD and high efficiency.

Start Up

Advanced start-up technology is used in the chip to meet the start-up time requirement. Low start-up current is designed for VCC can be charged above UVLO-Off with small current.

During the start-up, the capacitor voltage at CMP is pulled up quickly. The chip operates at short circuit and over current protection is set cycle-by-cycle until it senses the output voltage by FB up to about 1.3V. After that, the chips operates in close loop.

Loop Compensation

Output voltage is feedback to FB from the voltage divider with the auxiliary winding and compared with an internal reference (2.5V) at the positive input of the Error Amplifier (EA) for regulation.

The output of the EA is the CMP which can be compensated externally with a capacitor for loop stability. The system bandwidth shall be set below 20Hz to suppress the AC ripple of the line.

Constant Voltage Control

The chip samples the output voltage through the auxiliary winding, divides the voltage and compares it with the internal reference to form a closed loop, and then keeps the output voltage constant. The output voltage formula is:

$$V_{out} = V_{FB_REF} \times \frac{(R_{FBL} + R_{FBH})}{R_{FBL}} \times \frac{N_{sec}}{N_{aux}} - V_{SD} \qquad (1)$$

Where R_{FBL} is FB pull-down resistance, R_{FBH} is FB pull-up resistance, N_{aux} is the turns of the auxiliary winding, and V_{SD} is the voltage drop of rectifier diode on the secondary side. At the zero cross point of the secondary current, V_{SD} is almost equal to zero and can be neglected.

Active Power Factor Correction (APFC)

The duration of the turn on period (T_{on}) is generated by comparing an internal fixed saw-tooth wave with the voltage on the CMP pin. During steady state, the voltage on CMP pin is slowly varying due to the large external capacitor, therefor the T_{on} can be considered constant. In flyback operation, constant T_{on} and quasi-resonant operation provide input current shaping as sinusoidal as shown in Fig. 4 and achieving active power factor correction, thereby high power factor and low THD can be expected.

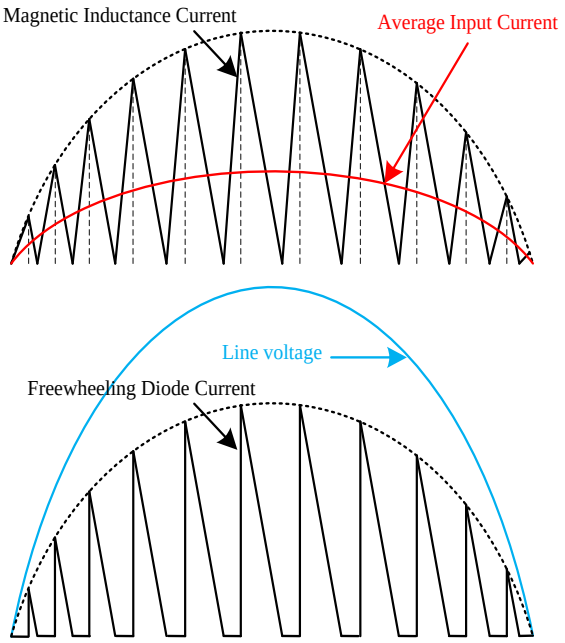


Fig. 4 Key wave forms of the CC controlled flyback with APFC

Quasi-resonant Operation

Dedicated zero crossing detection of the inductor current is designed for PWM turn-on signal, thereby the valley turn-on of the MOSFET and quasi-resonant operation can be achieved.

ZCD Blanking

Fig. 5 shows ZCD blanking time (T_{LEB_ZCD}) is set to avoid the fault trigger by the oscillation after the turn off.

And this blanking time is adjusted according to the CS peak to improve the EMC and efficiency performance.

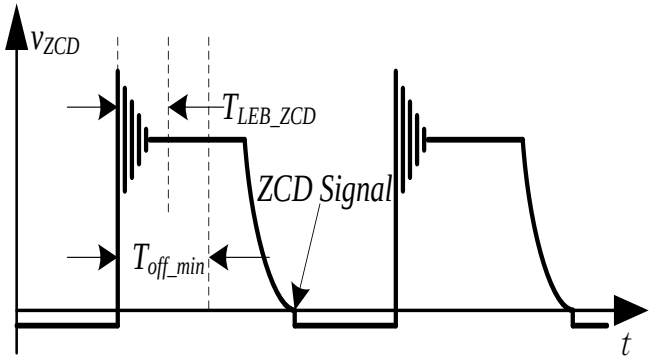


Fig. 5 ZCD blanking

Maximum Frequency Setting

Minimum off time (T_{off_min}) is set to limit the maximum switching frequency (f_{sw_max}), then the switching loss and EMC performance can be guaranteed. As shown in Fig. 6, PWM is triggered at the first valley after T_{off_min}.

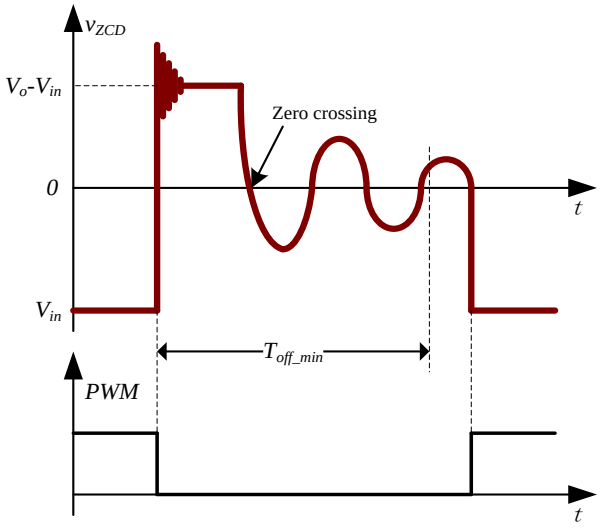


Fig. 6 Minimum off time

CS Blanking

Fig. 7 shows CS blanking time (T_{LEB_CS}) is set to avoid the fault trigger by the oscillation after the turn on.

Cycle by Cycle Current Limiting

When magnetic inductance saturation happens, CS pin voltage (V_{CS}) goes higher than regulation. However, CS voltage limit (V_{CS_OCP}) can help to stop PWM cycle by cycle to protect the MOSFET and the other power devices.

Short Circuit Protection (SCP)

When output is short, the FB voltage is low. If the voltage at FB pin is lower than FB short protection threshold V_{SCP_FB}, then the CS sampling OCP voltage V_{CS_OCP} is reduced to 0.4V (typical) to reduce the dissipation of the power.

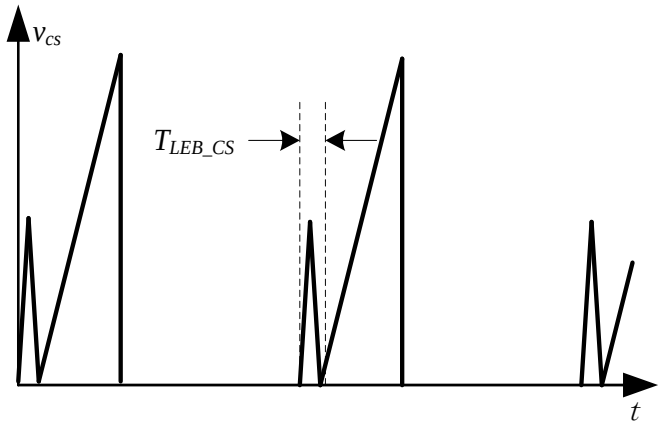


Fig. 7 CS blanking

Over Voltage Protection for Output

When the platform voltage detected by FB reaches the internally open circuit protection threshold V_{OVP_FB}, the system enters open circuit protection, the chip shuts down and enters into power on start-up sequence thereafter. The open circuit protection voltage is:

$$V_{OVP} = V_{OVP_FB} \times \frac{(R_{FBL} + R_{FBH})}{R_{FBL}} \times \frac{N_{sec}}{N_{aux}} - V_{SD} \qquad (2)$$

Where V_{OVP} is the over voltage protection point that needs to be set.

Over Load Protection (OLP)

Chip detects the output power and offers over load protection (OLP). If output power exceeds the power limit threshold (typical 1.3 times the full load), the chip reacts to turn off the power MOSFET. After 150ms (typical), the OLP stave can be reset.

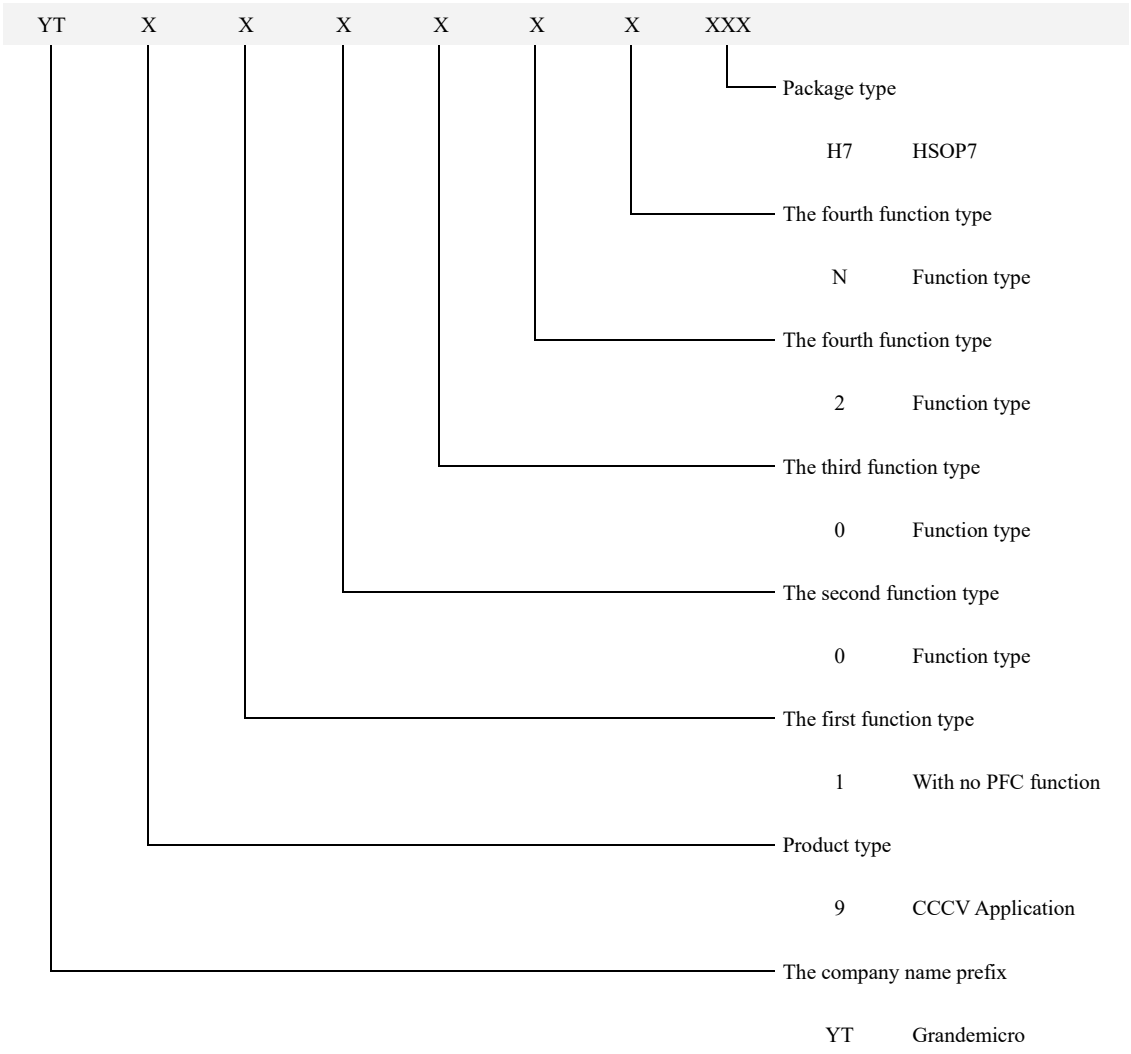
Over Temperature Protection (OTP)

When the junction temperature goes higher than the OTP Threshold, chip shuts down and keep the state until a new restart.

PCB Design Guide

1. Current sensing resistor (R_{cs}) shall be placed very close to the chip, minimize the loop from CS pin to R_{cs} and GND pin.
2. Separate the power ground and the signal ground. Please use star connection method to connect power ground, signal ground to GND pin.
3. Keep DRAIN1 floating with no connection.

Marking Rule

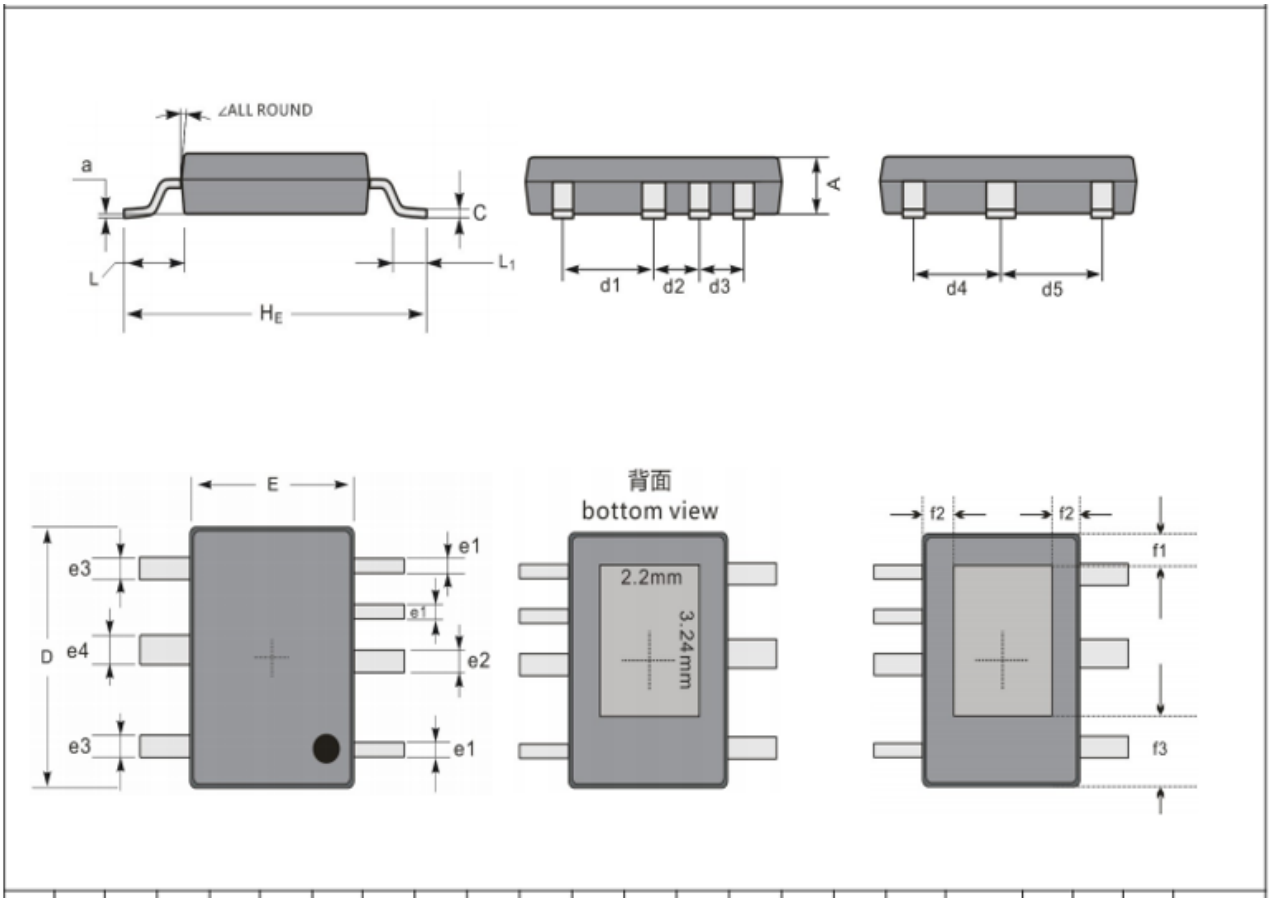


●	Part No. Date Code Special Note

Part No.	YT91002N
Date Code	P: Packing site; Y: Year; W: Week; S: Series; V: Version.
Special Note	Use obvious mark like “H”, “L” to distinguish the key features like CS or FB reference etc. if necessary.
●	The first pin marked

Package

HSOP7



Unit		A	C	D	E	HE	d1	d2	d3	d4	d5	e1	e2	e3	e4	L	L1	a	∠	f1	f2	f3
mm	max	1.25	0.22	6.4	4.1	6.1	2.56	1.38	1.32	2.28	2.78	0.45	0.56	0.60	0.85	1.15	0.8	0.2 (ref)	12°	0.71	0.9	2.35
	typ	1.15	0.20	6.2	3.9	6.0	2.51	1.33	1.27	2.23	2.73	0.40	0.51	0.55	0.80	1.05	/			0.66	0.85	2.3
	min	1.05	0.15	6.0	3.70	5.9	2.46	1.28	1.22	2.18	2.68	0.35	0.46	0.50	0.75	0.95	0.40			0.61	0.8	2.25
mil	max	49	9	252	161	240	101	54	52	90	109	18	22	24	33	45	31	8 (ref)		28	35	93
	typ	45	8	244	154	236	99	52	50	88	107	16	20	22	31	41	/			26	33	91
	min	41	6	236	146	232	97	50	48	86	106	14	18	20	30	37	16			24	31	89

Version History

Version	Date	Description
A0	May. 2025	Draft
A1	May. 2025	Released